

LESSON 7-5 • SESSION 1 • PRACTICE • FORM A

Plotting Points on the Coordinate Plane

Learning target – I can plot a point with rational coordinates and name the quadrant it lies in. • 6.NS.C.6c

REMEMBER

- An **ordered pair** (x, y) names a point. Start at the **origin** $(0, 0)$: x says right or left, then y says up or down.
- Signs name the **quadrant**: $(+, +)$ I, $(-, +)$ II, $(-, -)$ III, $(+, -)$ IV.
- A point with a 0 coordinate is **on an axis**, not in a quadrant.

WORKED EXAMPLE

Plot $(-5, -4)$. Name its quadrant.**Step 1** Start at the origin, $(0, 0)$.**Step 2** Move 5 units left, because $x = -5$.**Step 3** Move 4 units down, because $y = -4$.**Answer** $(-, -)$ is Quadrant III

WORDS coordinate plane (*plano de coordenadas*) • ordered pair (*par ordenado*) • origin (*origen*) • quadrant (*cuadrante*)

PRACTICE

show your work • label units

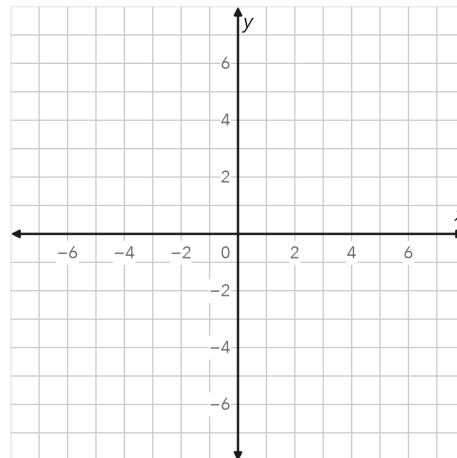
1. Name the quadrant or axis for each point.

a. $(-1, 6)$ _____b. $(4, 4)$ _____c. $(-8, -5)$ _____d. $(3, -9)$ _____e. $(0, -5)$ _____f. $(8, 0)$ _____

- 2.
- Park Map.**
- The fountain is at the origin. Gate
- $(6, 3)$
- , Pond
- $(-4, 8)$
- , Bench
- $(-7, -2)$
- , Swing Set
- $(5, -6)$
- .

Hint Move along the x -axis first.

- a. Plot and label all four spots.
- b. Quadrant: Gate ____ Pond ____ Bench ____ Swing Set ____
- c. Ordered pair for the fountain: _____



- 3.
- Build the Pair.**
- A point is 6 units left of the
- y
- axis and 11 units above the
- x
- axis. Write its ordered pair and quadrant.

Hint Left means x is negative.**Point** _____

- 4.
- Order Matters.**
- Name the quadrant of
- $(-5, 1)$
- and of
- $(1, -5)$
- . Are they the same point?

Hint Check the sign of x , then the sign of y .**Quadrants** _____

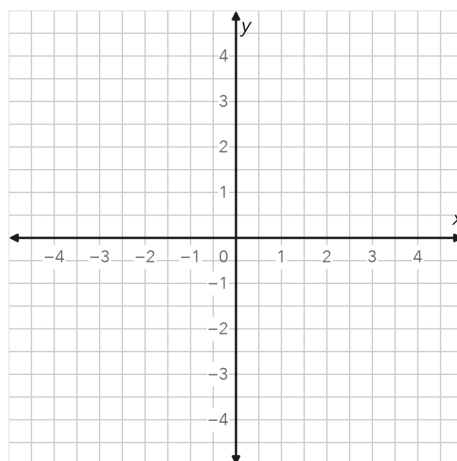
- 5.
- Same Digits.**
- Noor says
- $(-2, 6)$
- and
- $(6, -2)$
- name the same point because they use the same two numbers. Explain why she is wrong.

6. **Rational Coordinates.** Plot $(-3.5, 2.5)$ and $(4.5, -1.5)$.

Name each quadrant.

Hint Count 3 whole units, then half of the next.

Quadrants _____



7. **City Map.** The library is at the origin. The museum is at $(-4, -7)$. The stadium is at $(9, 1)$.

a. Quadrant: museum ____ stadium ____

b. Ordered pair for the library: _____

8. **Map Points.** A map shows a school at $(2.5, 3)$, a park at $(-2.5, 3)$, and a pool at $(-2.5, -3)$.

a. Name the quadrant of each place. _____

b. Which y-coordinate do two places share? _____

c. How many units straight down is the pool from the park? _____

FIND THE ERROR

explain the mistake, then fix it

9. **Dmitri's Work.** Dmitri plotted $(-5, 4)$ by moving down 5 units and then right 4 units. His point landed in Quadrant IV. Explain his mistake. Then describe the correct moves and quadrant.

TEST PRACTICE

choose the best answer

10. The point $(-4, 7)$ lies in –

A Quadrant I

B Quadrant II

C Quadrant III

D Quadrant IV

11. In the ordered pair (x, y) , the first number tells you to move –

A up or down

B right or left

C diagonally

D back to the origin

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

(0, 0) G	(11, -6) Q	(-6, 11) I	-3 X	(6, -11) Z	II O	6 N	3 I	y-axis R
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#1a	#1e	#3	#7b	#8b	#8c
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CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____